

Freight Rate Prediction

Model validation and submission report | September 2026

1. Task and data quality

Estimate posted load rates for 12,000 unlabeled November-December 2025 loads and a 31-day December fixed scenario. The labeled dataset contains 48,000 loads from January-October 2025. Training and submission IDs are unique and disjoint; no exact duplicate training feature rows or cross-dataset feature overlaps were found.

Training has 300 missing weights and 374 missing market-index values; submission inputs have 165 and 249 respectively. Numeric missing values are retained for native tree handling. Categorical missing values receive an explicit label. IDs and posted_rate are never predictors. Quote and market signals are assumed available because they are supplied as submission inputs; real-time availability should be confirmed.

2. Chronological validation

Stage	Training period / rows	Evaluation period / rows
Selection	Jan-Jun / 28,806	Jul-Aug / 9,671
Retrospective comparison	Jan-Aug / 38,477	Sep-Oct / 9,523
Final refit	Jan-Oct / 48,000	Nov-Dec / 12,000 (unlabeled)

On July-August, MAE selects model family and training length, with early stopping after 100 rounds without improvement (maximum 1,500 trees). Each candidate is then refitted through August at its fixed tree count and evaluated on September-October. Final models are refitted on all labeled rows. Both models use seed 42 and four threads.

Evaluation limitation: September-October was used in earlier exploratory scripts. These are retrospective results, not an untouched or unbiased final test. Hidden-label accuracy is calculated by Spotter. No November-December labels were available.

3. Features and modeling choices

Shared features: pickup, delivery, route, equipment, distance, weight, month, weekday, and annual sine/cosine terms. The full model additionally uses coordinates, market index, quote signal, and distance multiplied by quote signal. CatBoost handles categoricals directly; LightGBM mappings are learned from each training partition. Unseen LightGBM categories are missing values.

CatBoost uses depth 8, learning rate 0.03 and L2 regularization 5 with RMSE loss. LightGBM uses 31 leaves, minimum 30 child samples, learning rate 0.03, feature fraction 0.85 and L2 regularization 5. Selection uses MAE. Negative predictions are floored at \$0.01 for the output contract.

4. Retrospective accuracy: September-October

Task / model	Trees	MAE (\$)	RMSE (\$)	WAPE
full / baseline	0	246.02	711.80	10.28%
full / catboost	277	116.16	634.74	4.86%
full / lightgbm	137	146.27	643.30	6.11%
december / baseline	0	1,148.92	1,569.42	48.02%
december / catboost	96	159.66	647.21	6.67%
december / lightgbm	110	160.83	644.91	6.72%

Full baseline = distance x quote signal. December baseline = training median posted rate. MAE measures typical dollar error; RMSE emphasizes large errors; WAPE is sum absolute error divided by sum actual rates.

Selected full model: CatBoost, 277 trees. MAE \$116.16, a 52.8% reduction from the quote baseline. 96.4% of predictions fall within \$250; 90th-percentile absolute error is \$144.36. Mean prediction-minus-actual bias is \$-56.96.

Selected December model: LightGBM, 110 trees, based on lower July-August MAE (\$170.63 versus CatBoost \$181.53). Its September-October MAE is \$160.83. CatBoost performs slightly better in that later comparison; selection is retained to avoid choosing on the evaluation results.

5. Errors and interpretation

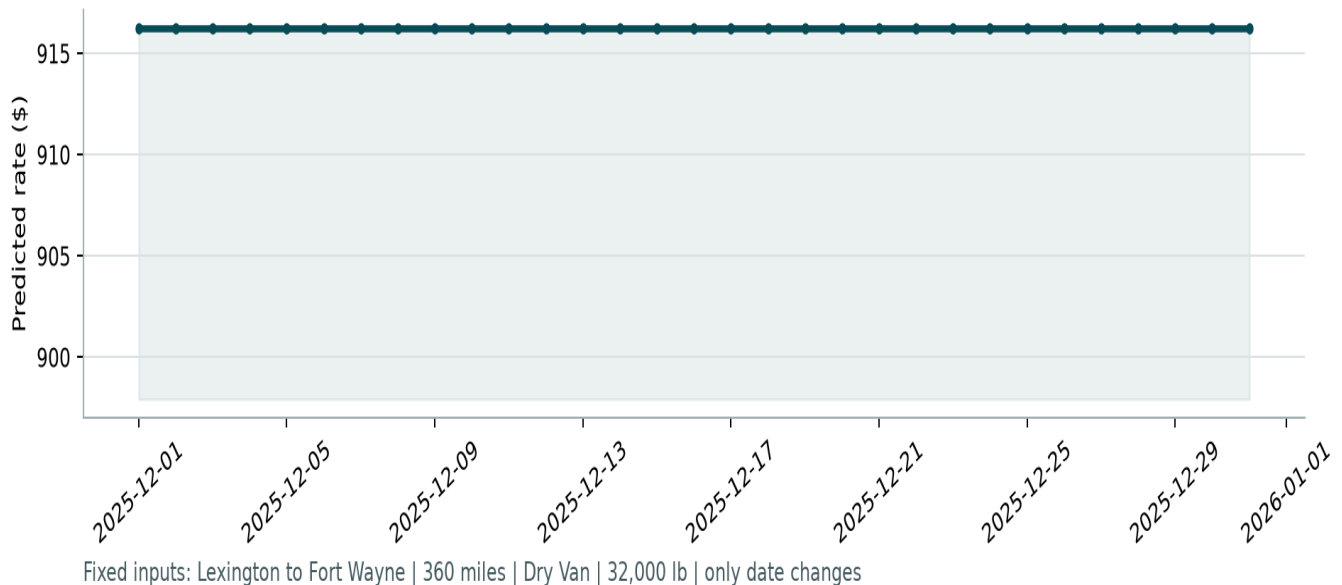
Segment	Loads	MAE (\$)	WAPE
equipment=Dry Van	5360	111.51	4.89%
equipment=Flatbed	1770	108.64	4.45%
equipment=Reefer	2393	132.16	5.07%
month=9	4670	105.37	4.38%
month=10	4853	126.55	5.32%

The largest 30 errors account for 25.9% of total absolute error. RMSE far exceeds MAE, indicating a heavy error tail. Large underpredictions warrant review of unusual loads and label quality; the available fields do not establish their cause. No difficult rows were removed from evaluation. Row-level predictions and worst-error extracts are supplied in artifacts/.

6. December fixed scenario

Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lb. Only date changes. Missing quote/market/coordinate fields are not fabricated: this scenario uses the reduced model trained and evaluated with the available fields.

Candidate: December 2025 Predicted Load Rate



Daily projected rates range from \$916.24 to \$916.24, with a mean of \$916.24. The chart is the supplied scorer's output. All 31 predictions are identical in this run: the selected model does not distinguish these dates for this scenario. No artificial daily variation has been added.

Forecast limitation: training contains less than one annual cycle and no November or December examples. Calendar features allow a date-dependent projection but do not establish December seasonality or holiday effects. Reduced-model retrospective accuracy is measured across all September-October loads, not exclusively this fixed lane. No confidence intervals or December accuracy claims are supported.

7. Reproduction and submission

Run `python freight.py all`, then `python score.py --predictions validation_predictions.csv --december-predictions december_chart_inputs.csv`. Run `python -m unittest discover -s tests -v` for focused input-contract checks. README documents Python 3.12, package installation, input paths, and outputs; requirements-tested.txt records the versions used.

Outputs include 12,000 ID-matched positive finite predictions, the completed 31-row December CSV, this report, and candidate_december.png. audit.json, metrics.csv and selection.json record the run. The supplied scorer checks file validity and creates the chart; it does not calculate hidden-label accuracy.

Before submission: verify additional requirements in Freight_Rate_ML_Assessment.pdf (unavailable during this review), upload the repository to GitHub with reviewer access, and record the accompanying 2-3 minute Loom walkthrough. The repository includes a recording script, not a fabricated Loom link.